

CSCE 658 RANDOMIZED ALGORITHMS · MODULE 09 · LECTURE NOTES

Markov Chains and Mixing

Random walks, and how long until they forget.

MODULE OUTCOMES

- Define a Markov chain and its stationary distribution.
- Explain mixing time and why it is the quantity of interest.
- Explain the spectral gap and conductance bounds.
- Explain MCMC and the Metropolis–Hastings construction.
- Diagnose a chain that is not mixing.

1 Chains, stationarity, and mixing

AN IRREDUCIBLE APERIODIC CHAIN CONVERGES TO A UNIQUE STATIONARY DISTRIBUTION

A Markov chain's next state depends only on the current state — CSCE 625 Module 10 §1's Markov property, without the actions and rewards that made it a decision process.

If the chain is *irreducible* (every state is reachable from every other) and *aperiodic* (there is no forced cycle length), then it has a unique stationary distribution π and converges to it from any starting state. That is the fundamental theorem, and both conditions are easy to check.

And for a simple random walk on an undirected graph, $\pi(v)$ is proportional to the degree of v — which is both simple to state and immediately useful: a random walk spends time proportional to degree, which is why PageRank needed a damping term and why degree-biased sampling is a hazard in network crawling.

So the question is never *whether* the chain converges but *how fast* — which is the mixing time, and it is the only quantity that matters for an algorithm built on the chain.

MIXING TIME IS THE QUANTITY, AND IT CAN BE ANYTHING

The mixing time is the number of steps until the chain's distribution is within ε of π in total variation distance, from the worst starting state.

And it varies enormously between chains: a random walk on a complete graph mixes in $O(1)$ steps; on a cycle it takes $\Theta(n^2)$; and on a dumbbell — two cliques joined by a single edge — it takes exponential time.

The dumbbell is the picture to hold. A bottleneck between two well-connected regions traps the walk on one side for a very long time, because escaping requires finding one particular edge among many. That is what slow mixing looks like geometrically, and §2's conductance bound makes the picture quantitative.

So 'it converges eventually' is not a useful statement. Every MCMC-based result depends on a mixing bound, and an unbounded one is an unsupported result (Module 13 §1) — which is a real and widespread problem in applied Bayesian work, where the mixing time is typically unknown and the diagnostics of §4 are the only available substitute.

2 Bounding the mixing time

SPECTRAL GAP: let the transition matrix's eigenvalues be $1 = \lambda_1 > \lambda_2 \geq \dots$. The spectral gap is $1 - \lambda_2$.

$$\text{mixing time} = O\left(\frac{1}{\text{gap}} \cdot \log(1/\epsilon)\right)$$

So a large gap means fast mixing. Exact, and frequently impossible to compute for the chain you actually have.

CONDUCTANCE (Cheeger's inequality): for a set S of states, the conductance of S is the probability of leaving S in one step, given that you are in S under stationarity. Φ is the minimum of that over all S with stationary measure at most $1/2$.

$$\text{gap} \geq \Phi^2 / 2 \quad (\text{and } \text{gap} \leq 2 \Phi)$$

So BOUNDING THE WORST BOTTLENECK BOUNDS THE MIXING TIME. This is the usable technique: find the tightest cut in the state space and measure how much probability flows across it.

THE DUMBBELL AGAIN: the cut between the two cliques has tiny conductance, so Φ is tiny, so the gap is tiny, so mixing is slow. The geometric picture and the algebraic bound agree exactly.

AND COUPLING is the third technique: run two copies of the chain from different starting states, show they can be coupled so as to meet quickly, and the meeting time bounds the mixing time. Frequently the easiest in practice.

Conductance is the usable bound of the three — it turns mixing time into a question about bottlenecks, which you can reason about geometrically and frequently answer by inspection. And coupling deserves its mention: it is what people actually use to prove mixing for specific chains, because exhibiting a coupling is often far easier than estimating either eigenvalues or the minimum conductance over all cuts.

3 Markov chain Monte Carlo

METROPOLIS–HASTINGS BUILDS A CHAIN WITH ANY STATIONARY DISTRIBUTION YOU WANT

Given a target distribution π known only up to a normalising constant, construct a Markov chain that converges to it. The normalising constant is precisely what you cannot compute — it is an integral over the whole space — and the construction does not need it.

Propose a move from the current state, then accept it with probability $\min(1, \pi(\text{new})/\pi(\text{old}))$ (times a proposal correction if the proposal is asymmetric) — and that is a *ratio*, in which the normalising constant cancels. That cancellation is the whole trick, and it is why the method exists.

The resulting chain is reversible with stationary distribution π , which follows from detailed balance and is a two-line verification — worth doing once.

And it is why MCMC is used everywhere: Bayesian posterior inference, statistical physics, and CSCE 647's path-space light transport all involve sampling from distributions known only up to a constant. The pattern is general: whenever you can evaluate a density up to proportionality and cannot normalise it, Metropolis–Hastings applies.

- **Bayesian posterior sampling** (CSCE 633 Module 09) — the posterior is known up to the evidence, which is exactly the intractable normalising constant, so this is the canonical application.
- **Metropolis light transport** (CSCE 647) — sampling light paths in proportion to their contribution to the image, which cannot be normalised because the total is what you are trying to compute.
- **Simulated annealing** (CSCE 669 Module 13 §1) — a Metropolis chain with a temperature schedule, where the target distribution concentrates on the optimum as the temperature falls.
- **Approximate counting. Counting perfect matchings or proper graph colourings reduces to sampling them nearly uniformly** — a deep and extremely useful connection, and the route to the only known approximation algorithms for the permanent.
- **And statistical physics**, where the method originated in 1953 and where the partition function is the uncomputable constant. **The counting-to-sampling reduction is the theoretically deepest item in this list** — approximate counting and approximate sampling are equivalent for self-reducible problems, which is Jerrum, Valiant and Vazirani's result and is what made approximate counting a field.

4 Diagnosis

AN UNMIXED CHAIN PRODUCES CONFIDENT, WRONG ANSWERS

A chain that has not mixed returns samples from whatever region of the space it started in — which look exactly like samples, have a mean and a variance and a tidy histogram, and are wrong.

And there is no internal signal. The chain does not know it has not mixed, and nothing in its output indicates the problem — which is CSCE 625 Module 09 §3's warning in its most practically consequential form.

So the diagnostics are all external: run multiple chains from widely different starting points and compare their distributions; compute the autocorrelation at increasing lags; monitor the effective sample size; compute the Gelman–Rubin statistic comparing within-chain and between-chain variance.

And the fix is usually a better proposal rather than a longer run. If the chain has a conductance bottleneck (§2), running it ten times longer does not help — the bottleneck is exponentially hard to cross — and a proposal that can cross it directly does. Which is why diagnosing slow mixing as a bottleneck rather than as insufficient patience is the important distinction.

- **Run at least four chains from different initialisations, always. Agreement between independently started chains is the only real evidence of mixing available, and a single chain is uninformative however long it ran.**
- **Report the effective sample size rather than the number of iterations — a million highly correlated samples may carry the information of fifty independent ones, and the iteration count is actively misleading.**
- **Discard a burn-in period, and be honest that its length is a guess informed by the diagnostics rather than derived from a mixing bound.**
- **Tune the proposal. An acceptance rate near 0.234 is asymptotically optimal for random-walk Metropolis in high dimensions — a genuinely useful constant, and a rate far above or below it indicates a proposal that is too timid or too bold.**
- **And prefer a better algorithm when one exists — Hamiltonian Monte Carlo and its adaptive variants mix dramatically better in high dimensions (CSCE 633) by using gradient information. 'Four chains and the effective sample size' is the minimum reporting standard, and it is directly analogous to CSCE 642 Module 12's five seeds — the same discipline, in a different stochastic setting.**

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Levin & Peres — Markov Chains and Mixing Times (free PDF)	Free from the authors, and the definitive treatment — chapters 1–7 and 12–13 cover §1 and §2, including coupling.

Resource	What it covers for this module
Mitzenmacher & Upfal — chapters 7 and 12	Markov chains and MCMC at this course's level , with the Metropolis construction derived.
Gelman et al. — Bayesian Data Analysis, 3rd edition (free PDF)	§4's diagnostics, free in full — including the Gelman–Rubin statistic and the effective sample size, by their originators.
Jerrum & Sinclair — The Markov chain Monte Carlo method (free survey)	§3's counting-to-sampling connection , and the conductance method applied to real chains.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Implement a random walk on a graph** and verify the stationary distribution is proportional to degree.
- 2 **Measure the mixing time empirically** for a complete graph, a cycle, and a dumbbell.
- 3 **Confirm the dumbbell is exponentially slow** by measuring crossing times.
- 4 **Compute the spectral gap** for a small chain and compare against the measured mixing time.
- 5 **Estimate the conductance** of the dumbbell's bottleneck cut and apply Cheeger.
- 6 **Construct a coupling** for a simple chain and bound the mixing time with it.
- 7 **Implement Metropolis–Hastings** for a distribution you know only up to a constant.
- 8 **Verify detailed balance** numerically.
- 9 **Run four chains from different starts** and compute the Gelman–Rubin statistic.
- 10 **Deliberately under-run a chain** and report the wrong answer it gives confidently.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 State the fundamental theorem and its two conditions.
- 2 What is the stationary distribution of a random walk on a graph, and what follows?
- 3 Define mixing time and give three chains with very different ones.
- 4 Describe the dumbbell and what it illustrates.
- 5 Give the spectral gap and conductance bounds, and say which is usable.
- 6 What is coupling, and why is it used in practice?
- 7 Explain why Metropolis–Hastings does not need the normalising constant.

- 8 Name five applications of MCMC.
- 9 What does the counting-to-sampling reduction establish?
- 10 Why are all mixing diagnostics external, and what is the minimum standard?